

Effector B-cell model: final Steps 1–3 summary

Main story: b6 controls long-time tumor outcome; b5 shifts the required b6 boundary.

Research question

Which effector B-cell parameter changes tumor dynamics?

Does the b6 threshold persist at long time?

Is the threshold robust to initial conditions?

Does b5 shift the b6 control boundary?

Final take-away: b6 is the strongest B-specific tumor-control parameter; b5 acts as a gatekeeper/modulator.

Workflow

Step 1 — dense b6 sweep to 10,000 days

Step 2 — initial-condition robustness

Step 3 — refined b5 × b6 phase diagram

Checks requested after the meeting are now addressed

Meeting suggestions

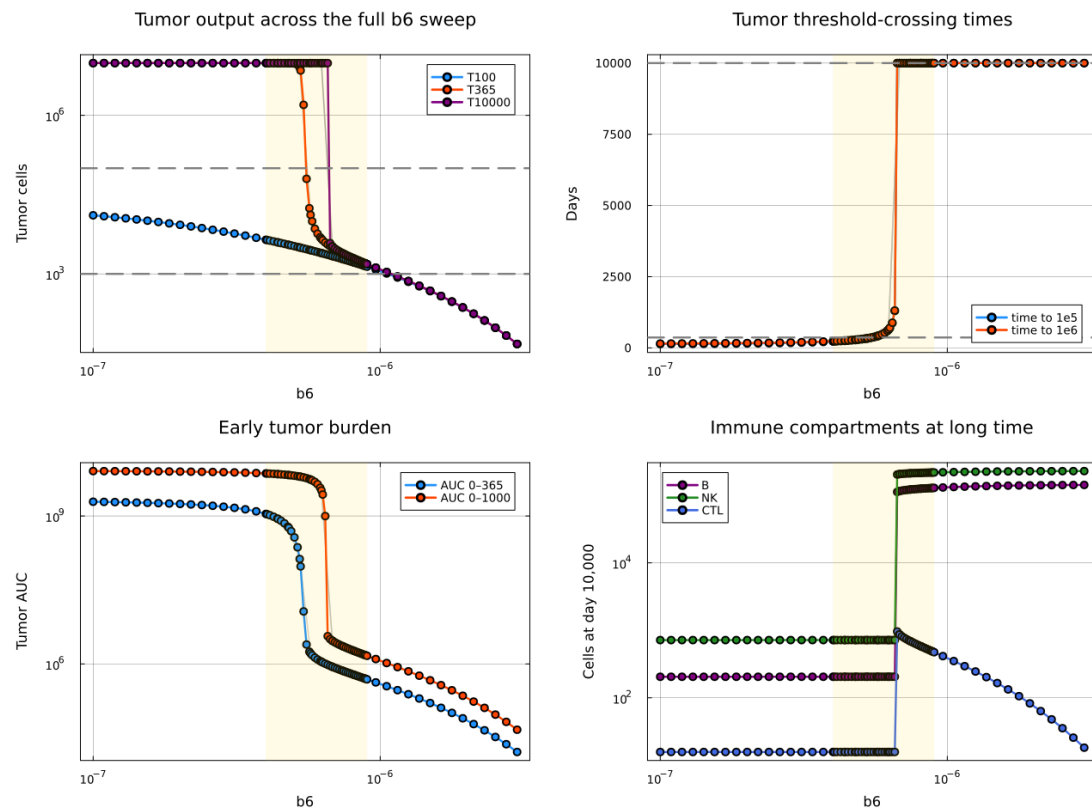
- Finer b6 grades
- Longer than 365 days
- Different initial conditions
- Baseline ICs listed
- B-cell low / NK effect state plots
- Cleaner heatmap / phase diagram
- Switch vs bifurcation question

Where addressed

- Step 1 dense sweep
- 10,000-day ODE runs
- Step 2 IC robustness
- Step 2 setup slide
- Step 2 supplement plots
- Step 3 b5 × b6 phase map
- Next: up/down sweep?

Step 1 finding: b6 has a long-time switch-like threshold

Dense 1D b6 sweep; direct CTL model; b5 fixed at 1e-4; baseline IC.



- T365 drops sharply near the original threshold region
- T10000 shows that some apparently controlled cases eventually regrow
- Long-time control requires a larger b_6 than day-365 control

The 10,000-day endpoint changes the interpretation: not all day-365 suppression is permanent control.

Key values from the dense CSV:

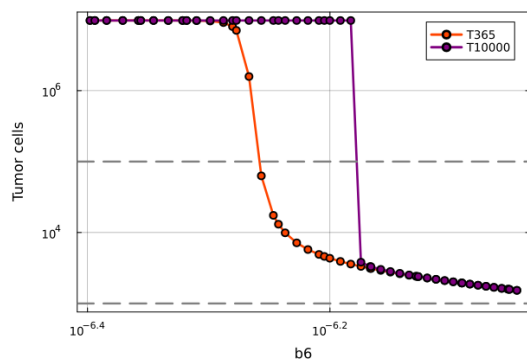
First T365 < 1e5: $b_6 \approx 5.54e-7$
First T10000 < 1e5: $b_6 \approx 6.69e-7$
First T10000 < 1e3: $b_6 \approx 1.15e-6$

Core result: b_6 is the main B-specific tumor-control parameter.

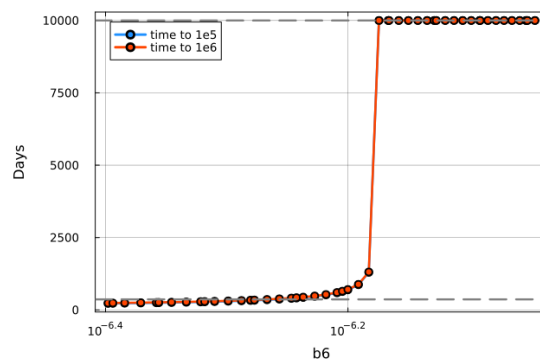
Step 1 zoom in: the transition differs by time horizon

The zoomed view separates finite-time delay from long-time control.

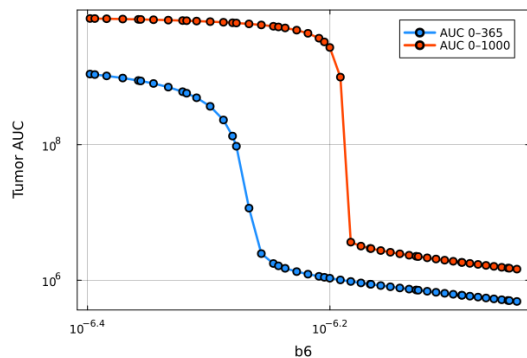
Zoomed threshold plot: T365 vs T10000



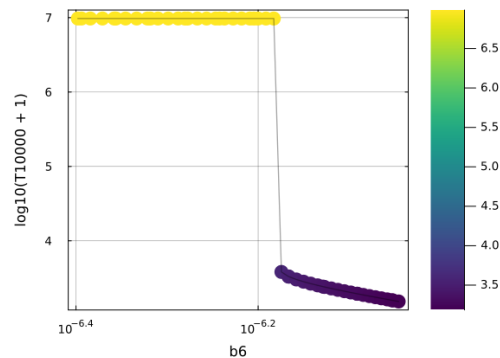
Zoomed threshold plot: crossing times



Zoomed threshold plot: early burden



Zoomed threshold plot: long-time state

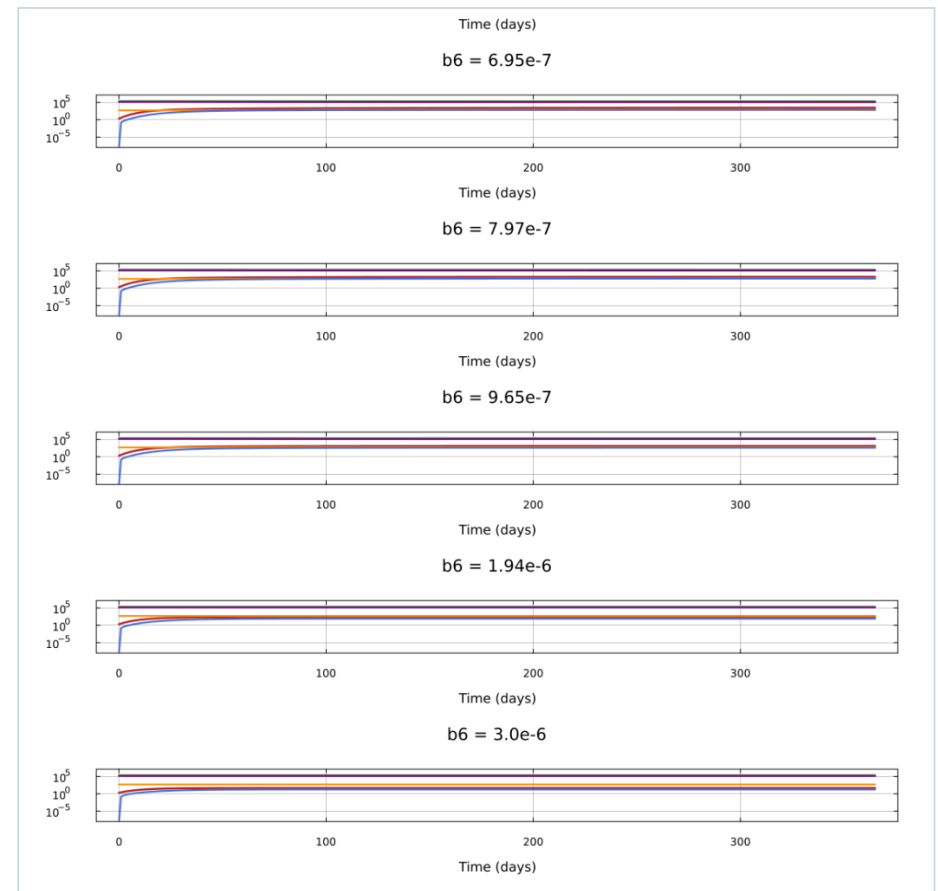
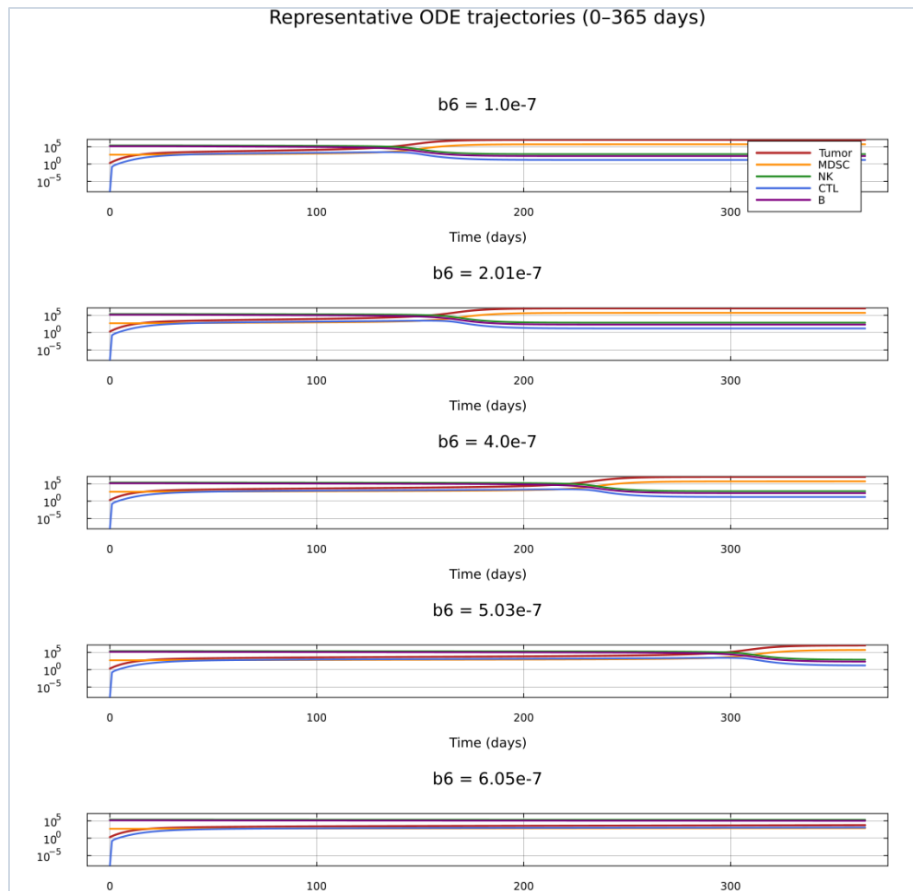


- At day 365, tumor burden falls earlier in the b_6 sweep
- Some regrow by day 10,000.
- Still, long-time control needs larger b_6

Interpretation: b_6 creates both a finite-time suppression threshold and a stricter long-time control threshold.

Step 1: Representative 0-365 day trajectories show early b6-dependent suppression

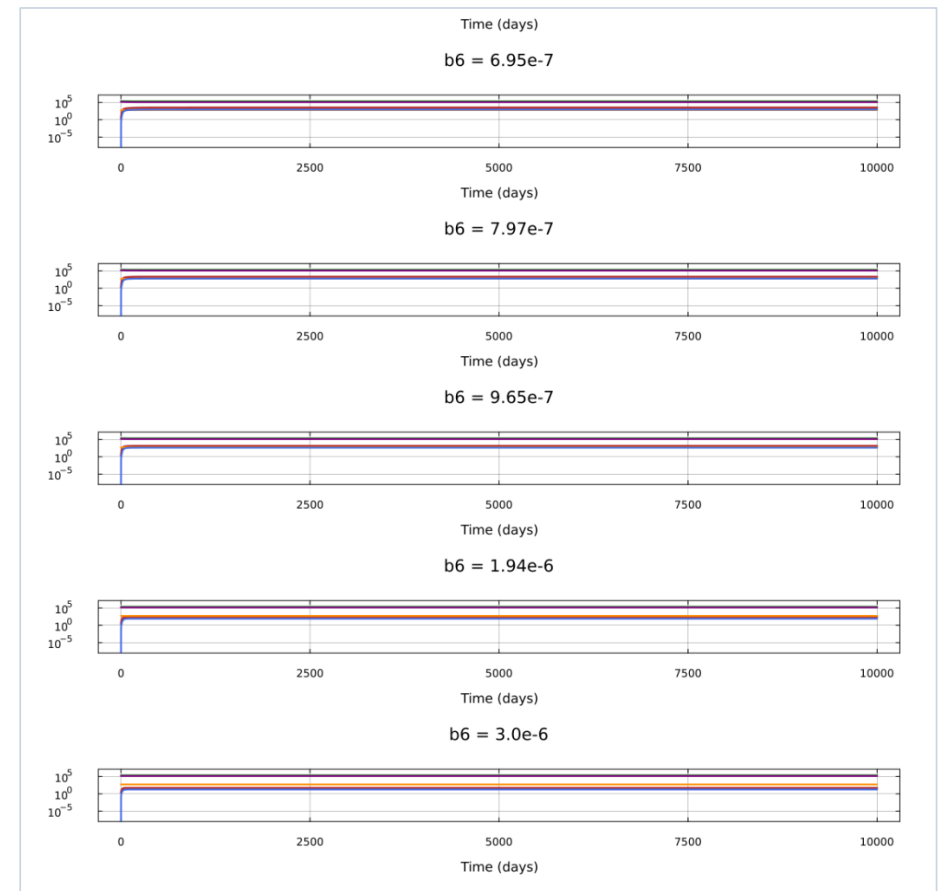
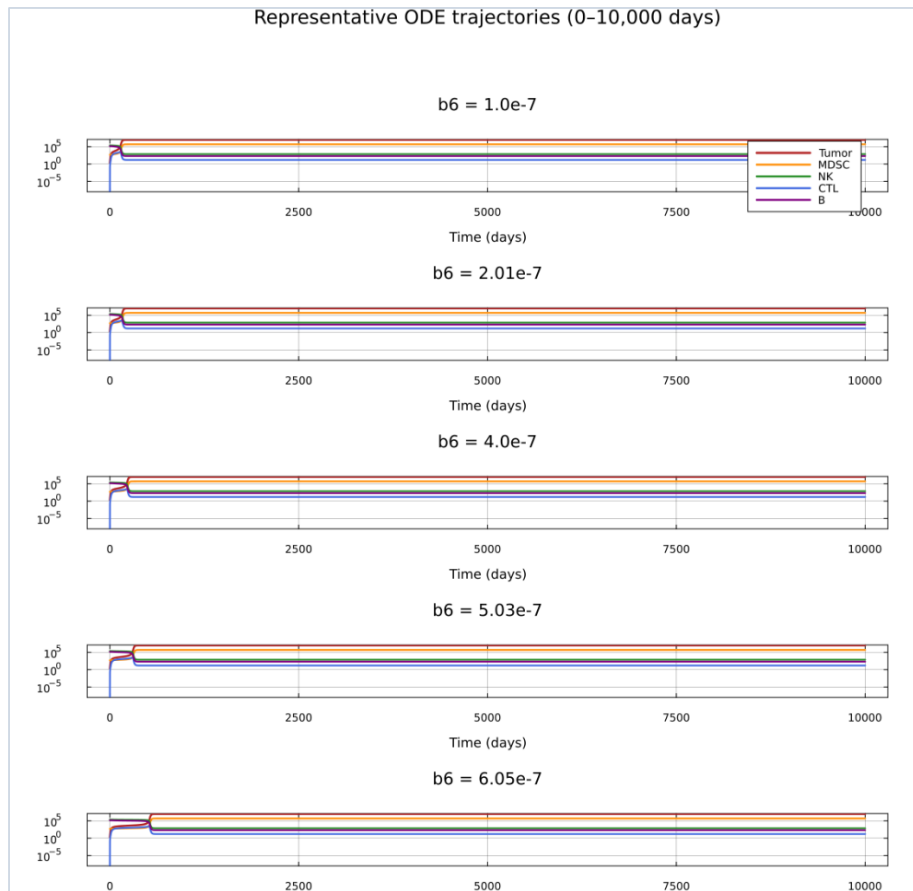
Short-time view preserves the early transient behavior



Early view: low b_6 remains tumor-dominant; near-threshold b_6 strongly delays tumor expansion.

Step 1: 10,000-day trajectories separate transient delay from long-time control

Long-time view checks whether day-365 suppression persists

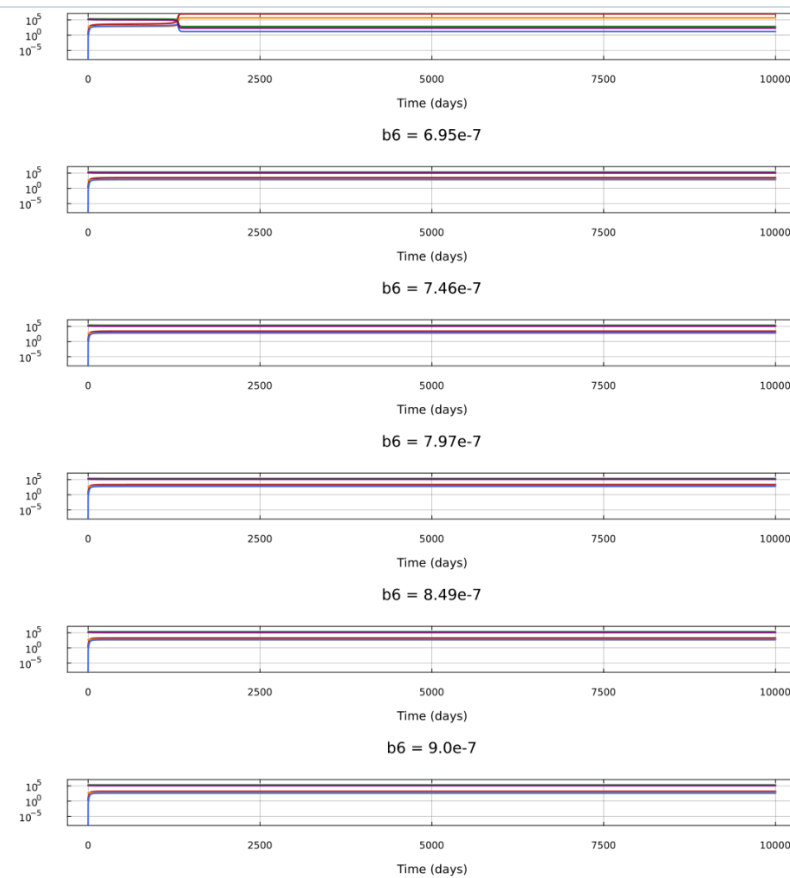
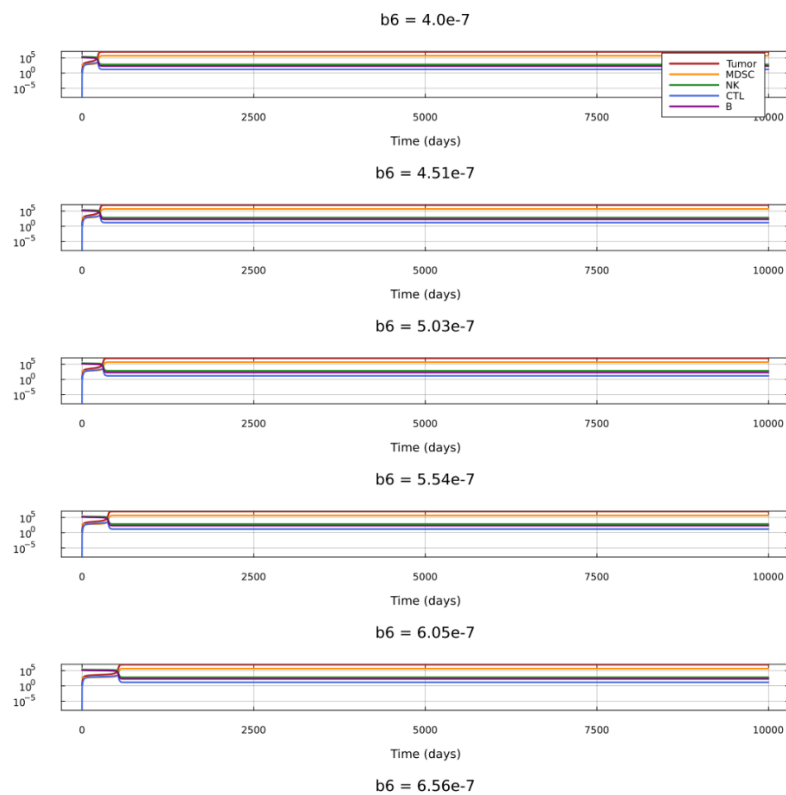


Long-time view: some mid-threshold cases are delayed but not permanently controlled; higher b_6 remains low long-term.

Step 1 : Threshold-region trajectories resolve the fine-grid transition

Fine b6 grid: 4e-7 to 9e-7, simulated to 10,000 days

Threshold-region ODE trajectories (fine b6 grid, 0-10,000 days)



Finer $b6$ grade is visible near the transition, and shows intermediate behavior, not a single coarse point; Higher values cross toward controlled regime.

Step 2 setup: initial-condition robustness

Question: is the b6 threshold only a baseline-IC artifact?

Change one initial condition at a time.
One-at-a-time perturbation for Tumor, MDSC, NK, CTL, and B.
Multipliers: 0.1x, 0.2x, 0.5x, 1x, 2x, 5x, 10x.
Readout: b6 threshold for T(10000) below 10^5 and 10^3 .

Supplement includes all-variable state plots for low NK(0) and low B(0).

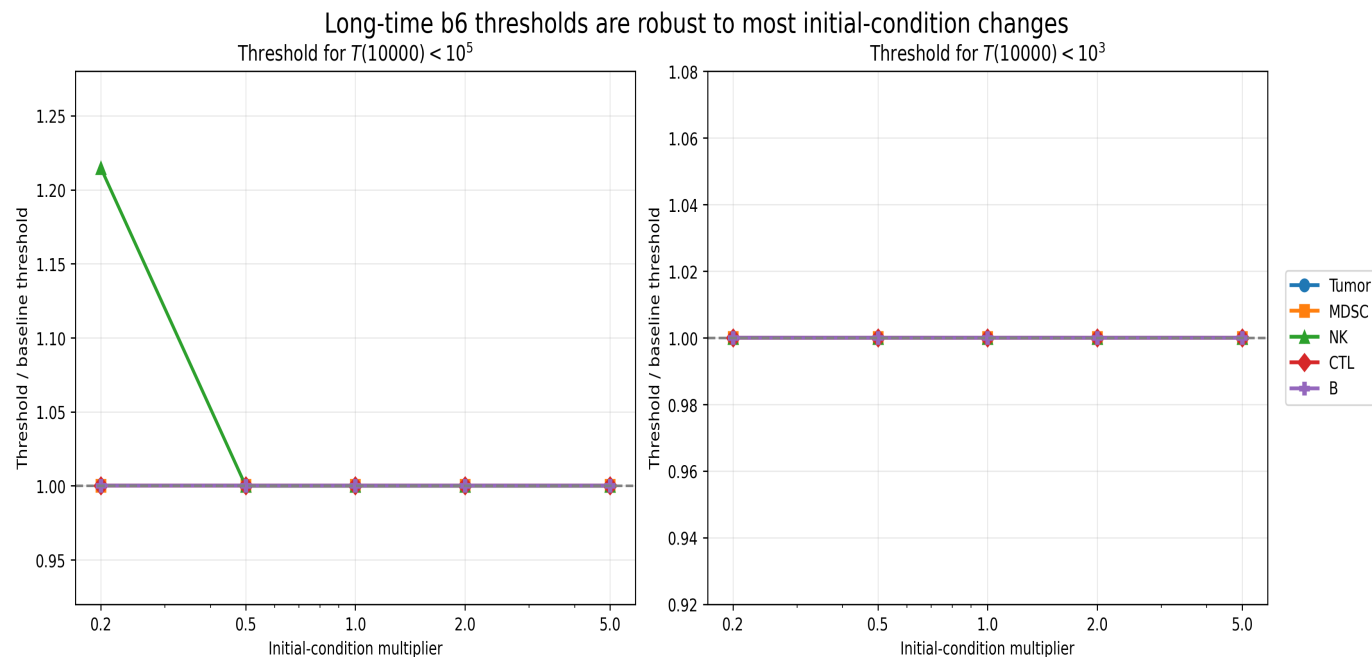
IC: initial condition

Baseline initial conditions

Variable	Baseline IC
Tumor	2.0
MDSC	500.0
NK	228758.170
CTL	0.0
B	142857.143

Step 2 : most IC changes barely move the long-time b6 threshold

Only low NK(0) clearly shifts the threshold.



$T(10000)$ = tumor cell count (size) after 10000 days

$T < 1e5$: loose threshold

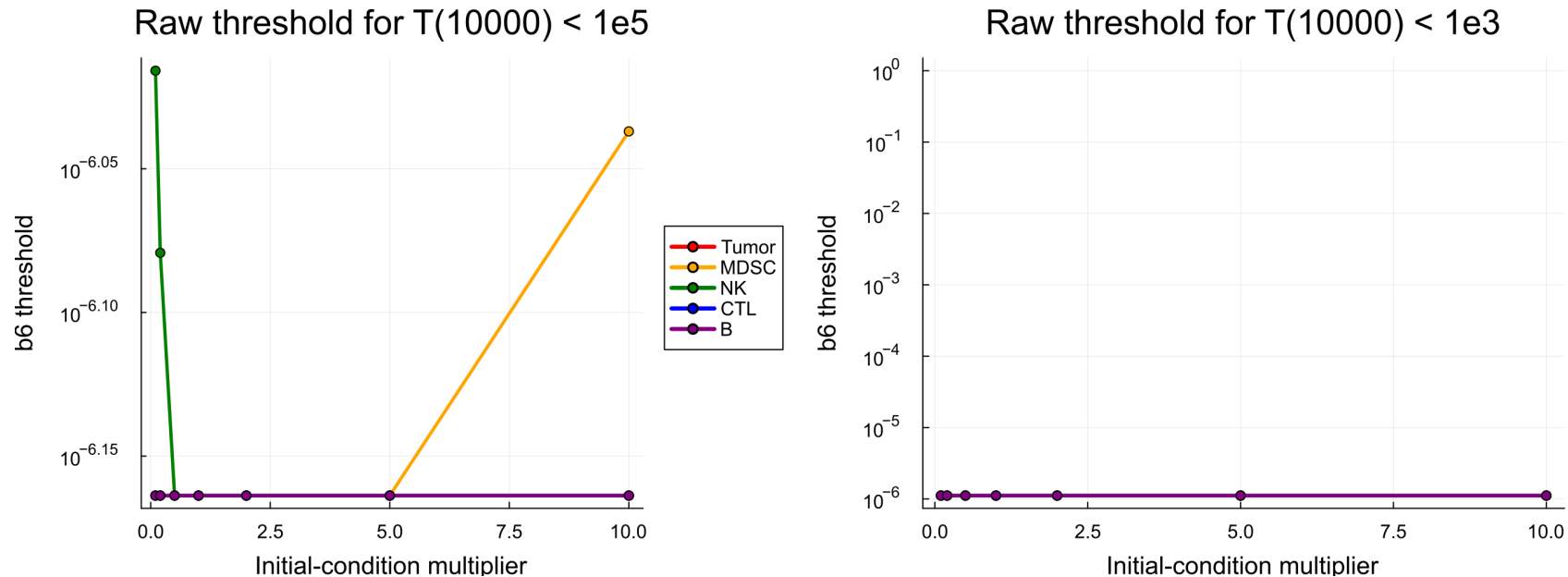
$T < 1e3$: strict threshold

Readout

- For $T(10000) < 1e5$, almost all IC perturbations stay near the same b6 threshold.
- Only $NK(0) = 0.2x$ baseline clearly shifts the loose threshold upward
- For $T(10000) < 1e3$, the stricter threshold is essentially unchanged across tested ICs.

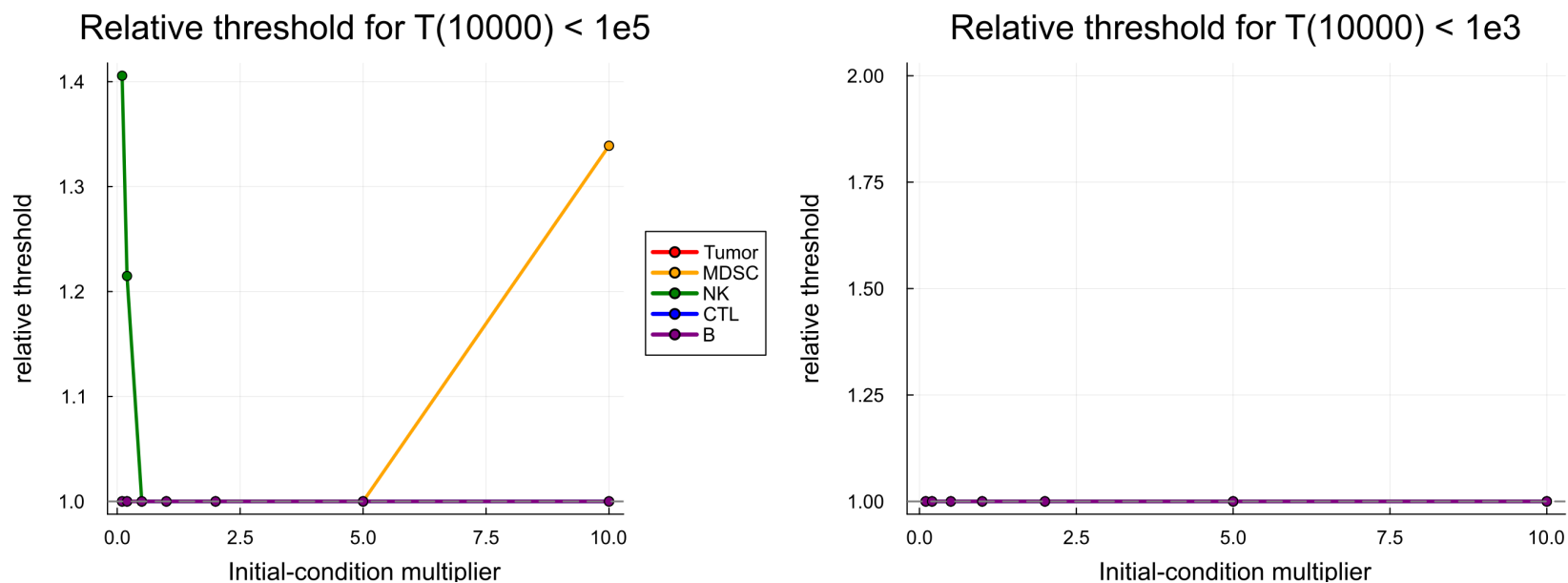
Interpretation: the long-time threshold is robust; low NK mainly affects how much b6 is needed for partial control.

Step 2 : Expanded IC check supports the same robustness trend



- Added 0.1 and 10× checks.
- Loose threshold shows extra sensitivity for low NK(0) and high MDSC(0).
- Stricter threshold remains stable.
- Low B(0) does not overturn the main conclusion.

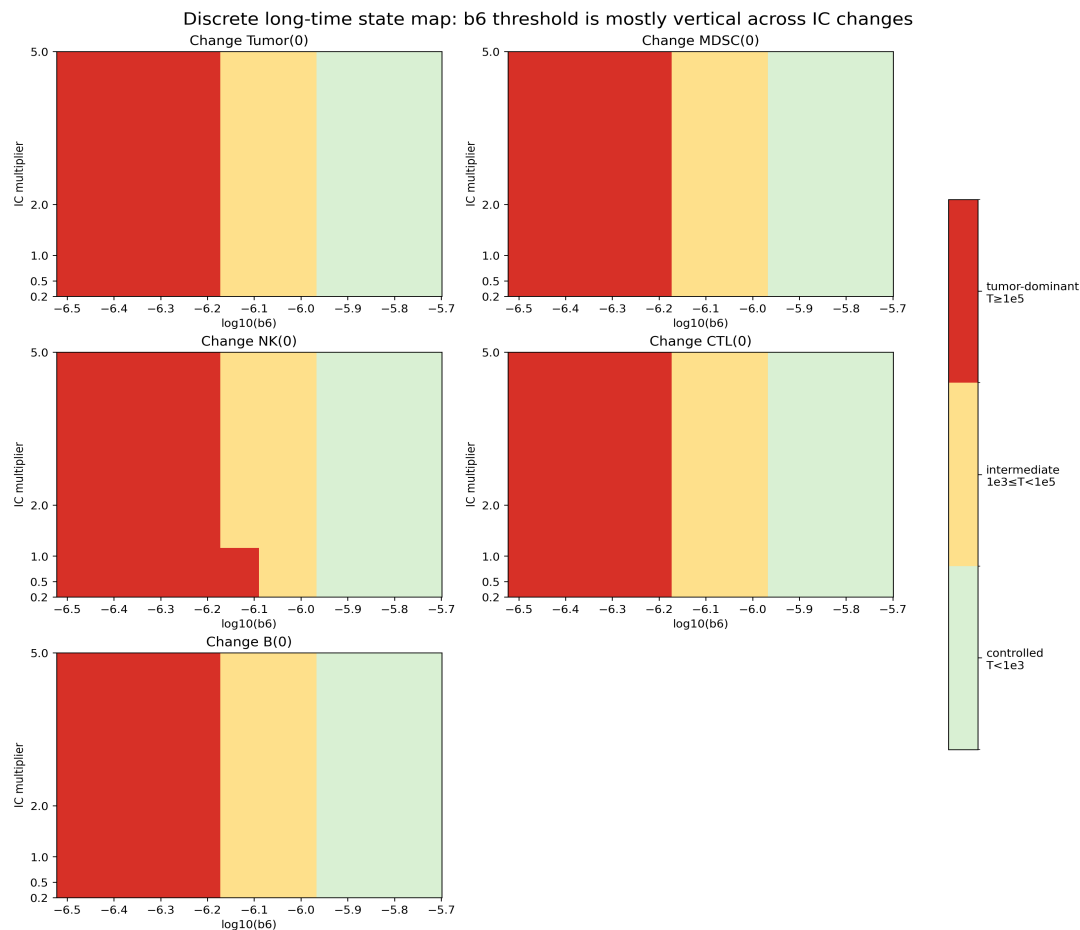
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Step 2 : the phase boundary stays almost vertical across IC changes

Phase-style heatmap: controlled / intermediate / tumor-dominant regimes



What the heatmap shows

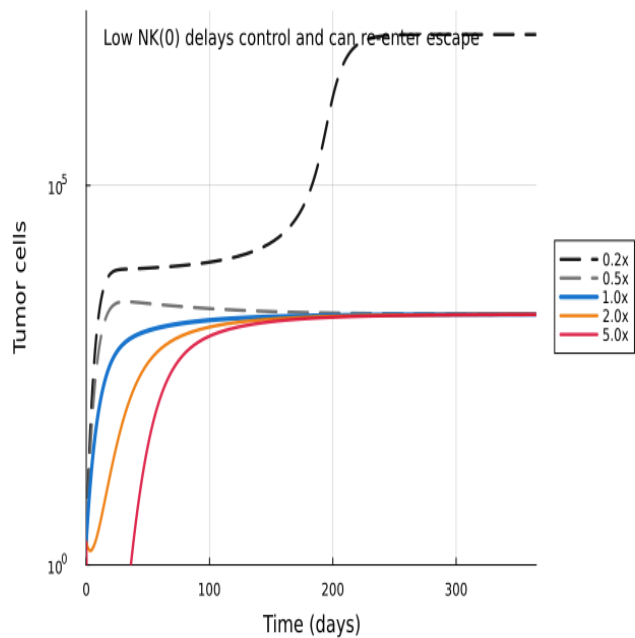
- Across Tumor(0), MDSC(0), CTL(0), and B(0), the regime boundary barely shifts.
- The main exception is low NK(0), which modestly pushes the boundary to the right.

Takeaway: b6 controls the regime, while most IC changes do not.

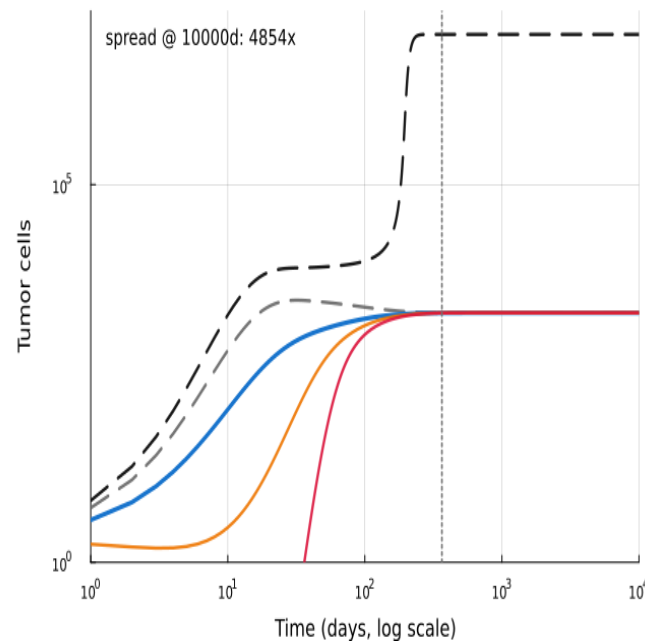
Step 2: near the threshold, low NK(0) is the only clearly visible sensitivity

Representative trajectories at $b_6 = 8e-7$

Step 2 highlight: NK(0) is the only IC perturbation with a strong effect near threshold
NK(0) perturbation: early-time divergence



NK(0) perturbation: long-time fate



Varying Tumor(0), MDSC(0), CTL(0), or B(0) gives very similar long-time tumor trajectories

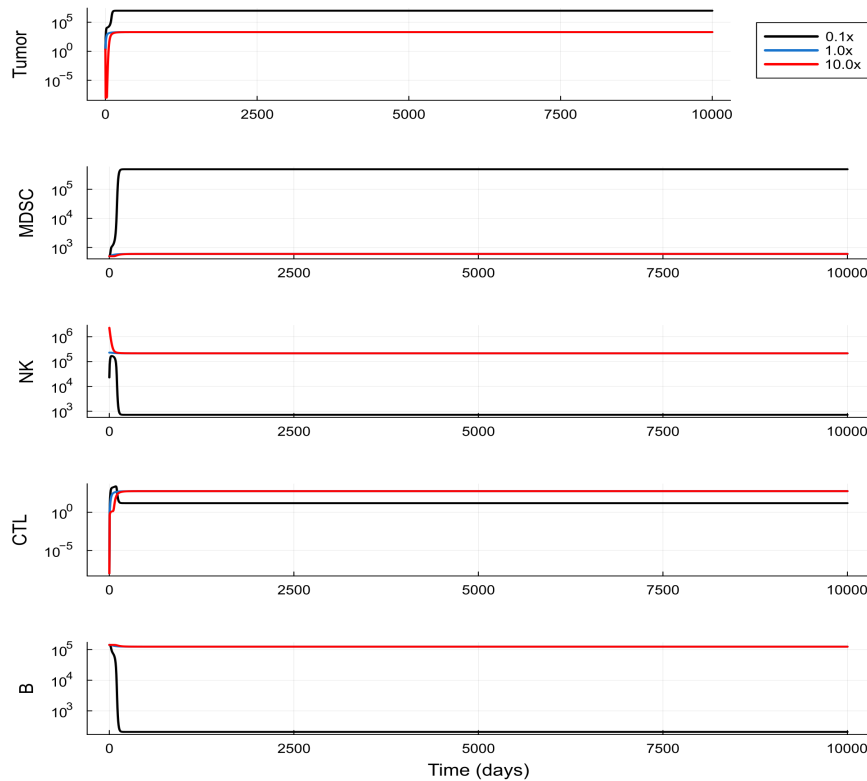
Reducing NK(0) makes tumor control harder near the threshold

NK is the main IC sensitivity to carry forward.

Step 2: low NK(0) is the only clear sensitivity near the threshold

Full-state trajectories at $b_6 = 8e-7$

Supplementary state trajectories: vary NK(0), $b_6=8.0e-7$



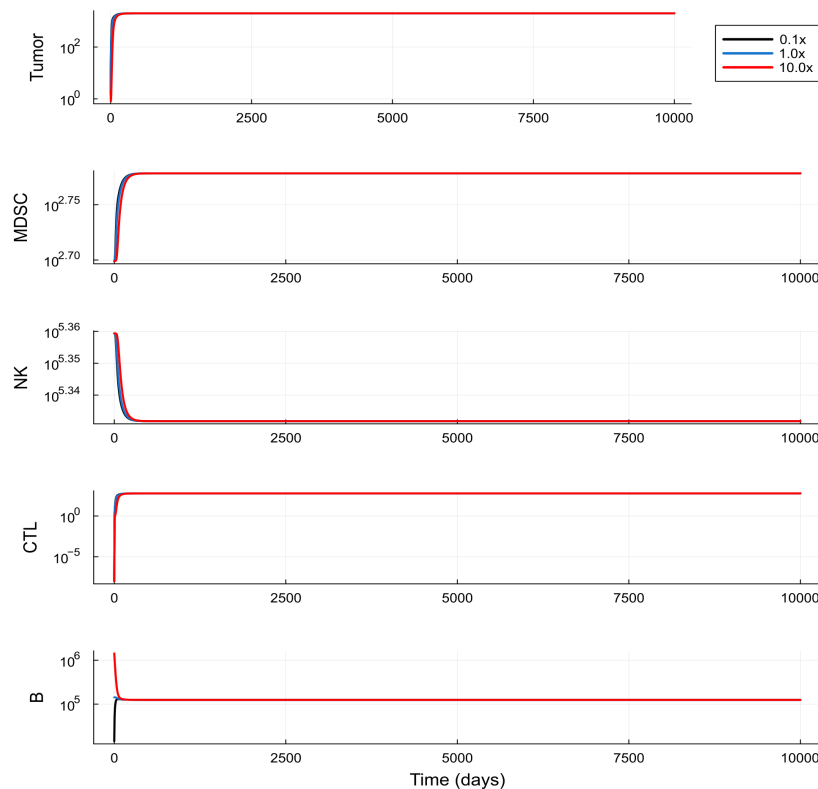
Why NK matters

- Reducing NK(0) makes tumor control harder near the threshold.
- Other IC changes do not produce a comparably strong long-time effect.

Supplementary check: lowering $B(0)$ does not change the threshold story

Expanded IC range includes 0.1x and 10x

Supplementary state trajectories: vary $B(0)$, $b_6=8.0e-7$



Why this was added

- Low $B(0)$ was tested Even with 0.1x and 10x checks, the qualitative threshold story stays the same.
- So the switch-like behavior is not simply driven by a large baseline B compartment.

Expanded 0.1x-10x checks support the same main conclusion.

Step 2 Summary and next steps

Step 2 asks whether the long-time b6 switch survives IC perturbations

What Step 2 establishes

- The long-time b6 threshold persists under most one-at-a-time IC changes.
- The stricter control threshold is especially stable.
- Low NK(0) is the only clear sensitivity in the tested range.
- Expanded 0.1x-10x checks do not change the main conclusion.

Potential next steps

- Refine the long-time b5 × b6 phase diagram.
- Run up-sweep / down-sweep continuation to test switch vs bifurcation-like behavior.
- If more IC testing is needed, prioritize more extreme or combined NK-focused conditions.
- Return to SDE only after the long-time ODE transition is fully characterized.

Bottom line: keep the direct CTL model, treat b6 as the main parameter, and use NK-focused checks to probe the transition further.

Step 3: $b_5 \times b_6$ phase diagram

Goal: test whether MDSC suppression of B (b_5) shifts the long-time b_6 control threshold

Fixed model

10,000 days

$b_5 \times b_6$ grid

Question

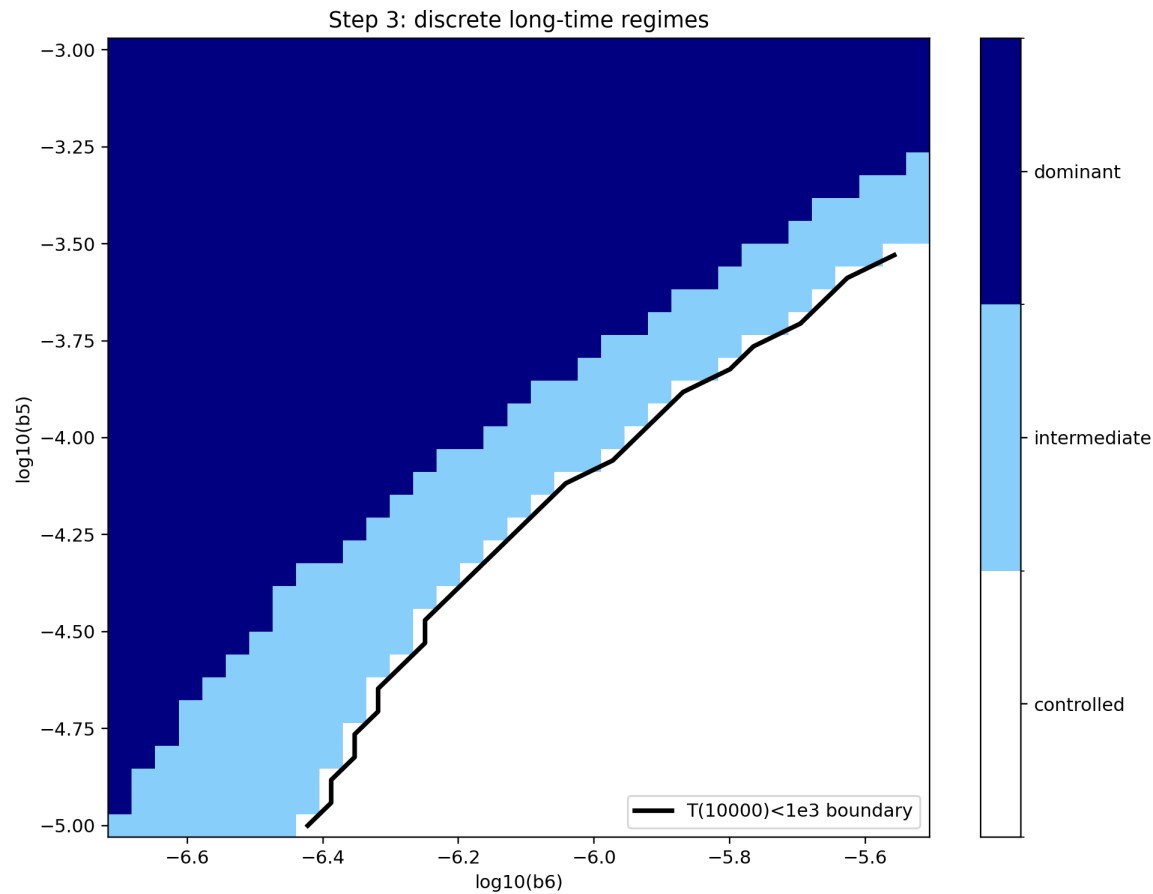
Does stronger b_5 make it harder for b_6 to control tumor growth?

- Direct CTL formulation kept fixed.
- Baseline initial condition kept fixed.
- Readout: long-time tumor burden $T(10000)$.
- Main visual: phase-style $b_5 \times b_6$ map.

Expected interpretation

- b_6 = B anti-tumor strength.
- b_5 = MDSC suppression of B.
- If b_5 increases, B is suppressed more.
- Therefore, larger b_6 may be needed for control.

Finding 1: $b_5 \times b_6$ forms a clear long-time phase boundary

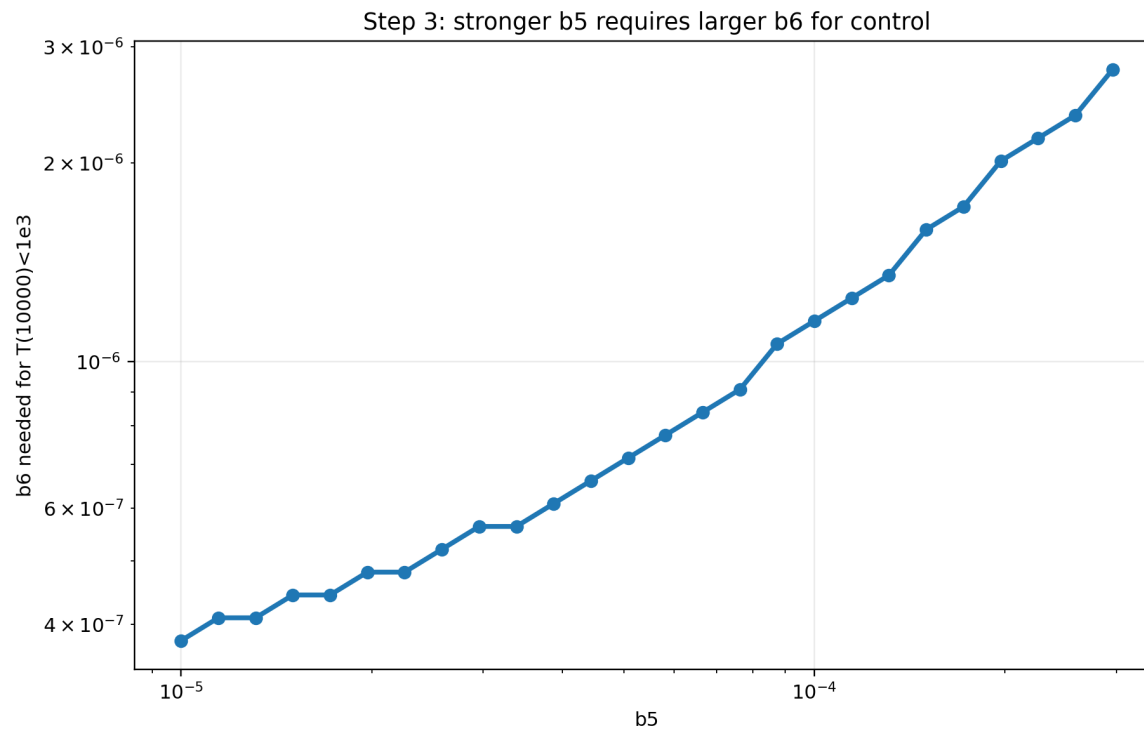


- Discrete regimes summarize long-time tumor outcome.
- The boundary is diagonal, not random.
- Higher b_5 shifts the control region toward higher b_6 .
- This supports the “ b_5 gates b_6 ” interpretation.

Main message

b_5 changes how much b_6 is needed for long-time tumor control.

Finding 2: stronger b5 requires larger b6 for control

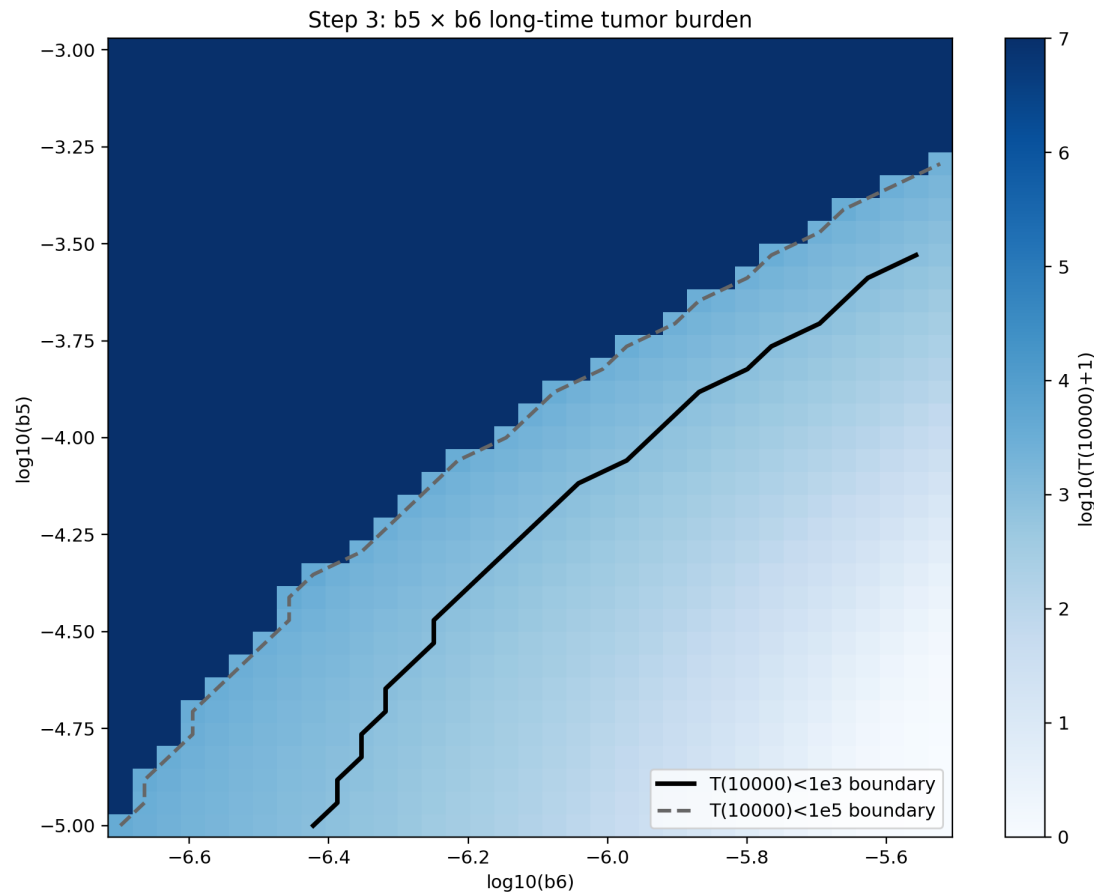


- Each point = smallest b_6 with $T(10000) < 10^3$.
- Boundary increases monotonically with b_5 .
- Low b_5 : smaller b_6 is enough for control.
- High b_5 : control requires larger b_6 .
- Very high b_5 may exceed the scanned b_6 range.

Quantified gatekeeper effect

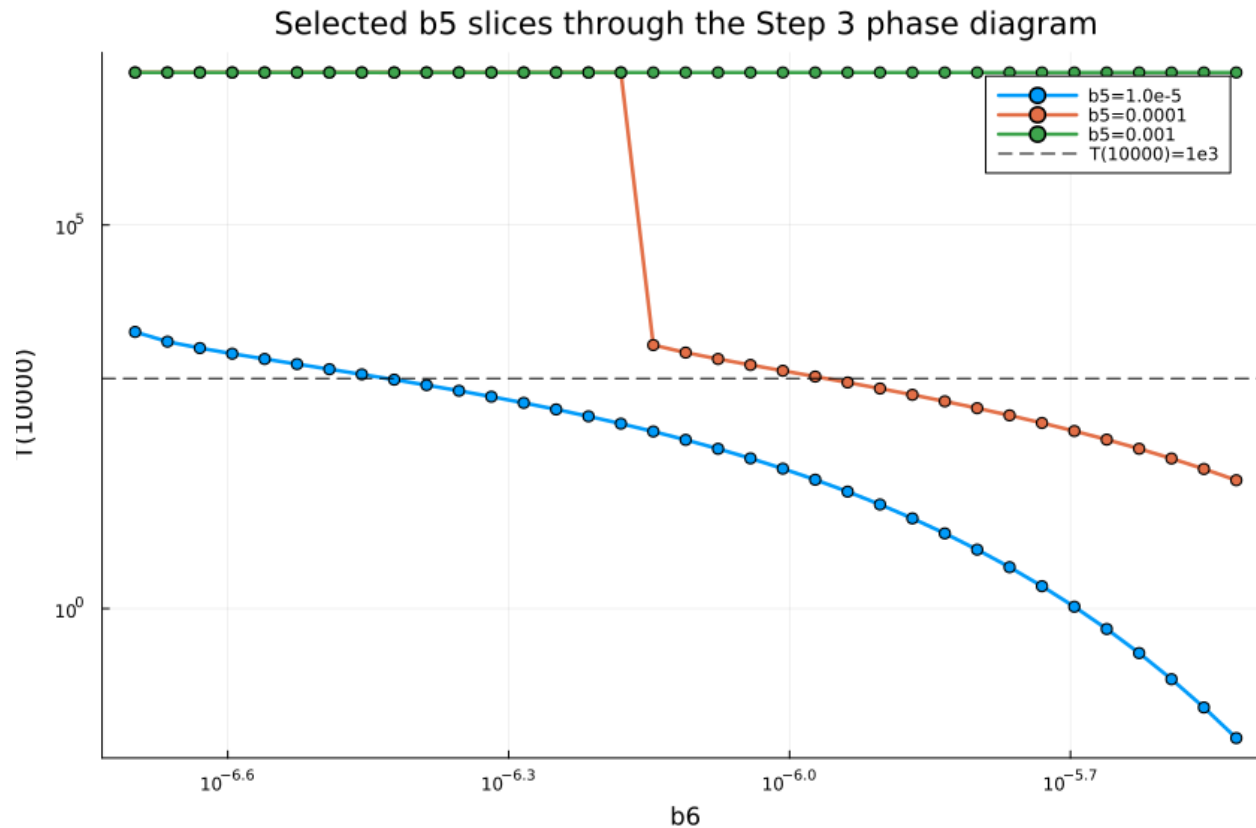
This plot turns the heatmap boundary into a simple trend.

Finding 3: log tumor burden confirms the same structure



- Log scale avoids raw-zero color artifacts.
- Black line: strict control boundary $T(10000) < 10^3$.
- Dashed line: looser low-tumor boundary $T(10000) < 10^5$.
- Both boundaries move upward with b_5 .

Finding 4: b5 slices show the transition in a simpler 1D view



- Low b5 slice: tumor can fall below the control line.
- Medium b5 slice: threshold shifts to larger b6.
- High b5 slice: tumor stays high in the scanned range.
- This cross-section supports the phase map and boundary plot.

Interpretation

b5 suppresses B, so a larger b6 is needed to overcome that suppression.

Step 3 summary: b5 is a gatekeeper for the b6 effect

Main findings

- Step 3 confirms a structured $b5 \times b6$ long-time landscape.
- Higher $b5$ pushes the control boundary to larger $b6$.
- This supports the mechanism: MDSC $\rightarrow$ B reduces B-mediated tumor control.
- The discrete phase map is the clearest main figure.

Next steps?

- Next modeling question: up-sweep / down-sweep to test switch vs bifurcation-like behavior.
- Do we need to return to SDE?